

Processing Audio Live on a Raspberry Pi

AKA: Real-time

Goal

- ▶ Electric guitar as source
 - ▶ Or any other physical instrument producing analogue signals
 - ▶ E.g: vocals, theremins, “vintage” electronic instruments



- ▶ Sound quality
 - ▶ On board sound processing poor quality up to Taspberry Pi 5
 - ▶ Raspberry Pi dispences with any audio interface
 - ▶ Need input and output



Qzn3t Pedal

- ▶ Simulated guitar effects pedal
- ▶ A simulated guitar pedal for live use

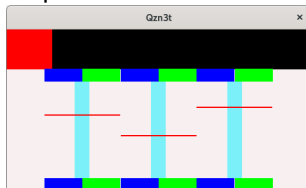


Figure 1: Screen shot

Inspiration: mod-ui



Figure 2: mod-ui screenshot

Uses

- ▶ LV2 simulators
- ▶ Jack Audio pipes
- ▶ Some other stuff I care not about

Problems

I want to:

- ▶ Use a MIDI Pedal live,
- ▶ I to select between simulated pedals
 - ▶ Not just LV2



Figure 3: SINCO Pedal

`mod-ui`

- ▶ Switching time of hundreds of milliseconds
 - ▶ Python all the way up and all the way down
- ▶ Only uses LV2. Jack can carry any audio
- ▶ Not very portable
 - ▶ Python 3.10 is not compatible with Python 3.11
 - ▶ Can work around but a very loud warning that this will not keep working with updated OS

(Activate on release - Aghhh....)

Solution

- ▶ Set up all the simulators in memory
- ▶ Connect the “pedalboards” with the Jack audio pipes between them
 - ▶ Do not set them up with I/O to the hardware devices
- ▶ Write a MIDI controller to change Jack audio pipes to enable I/O

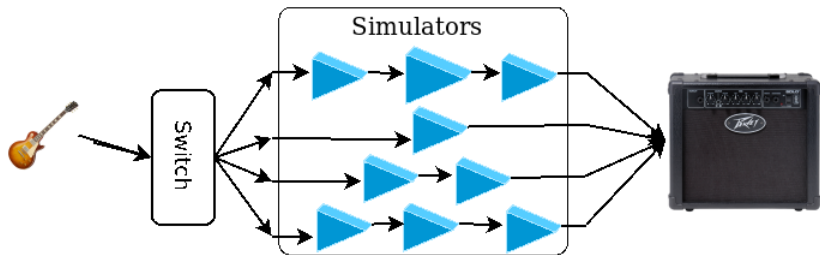


Figure 4: MIDI Selecting Simulators

More Problems

- ▶ Pison sound hardware feels delicate
 - ▶ The case is inadequate
 - ▶ Power supply will not connect if front plate attached
 - ▶ Without front plate female jacks for audio I/O are vulnerable to metal fatigue of solder joints





Figure 5: Cigar box solution

Not confidence inspiring nor attractive

Solution: Mark II

- Stay with Raspberry Pi 5



- Use a different case
 - Regrets: Should have used official 7" screen 3.5" too small



- ▶ Use a USB “sound card”
 - ▶ Cheap and cheerful (irritating I/O female sockets - 3mm plugs)



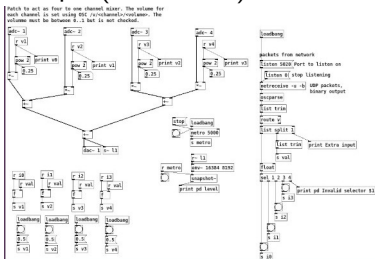
- ▶ Lowest rated latency (not cheap at all ~\$300)



There are many choices....

Not Finished Yet

- ▶ Adjusting settings interrupts creative process
- ▶ Most important setting is volume
 - ▶ Difficult to estimate subjective and relative volume of electric guitar on its own - as opposed to “full noise”
- ▶ Use pd (Pure Data) to make simple mixer to control volume



(This mixer only ever has input on one channel. It does not correctly mix audio data for more than one channel)

Complications

- ▶ Out of order: Mixer $\Rightarrow$ Touch screen - for control
- ▶ Use OSC to control mixer
 - ▶ A simple protocol
 - ▶ Supported by pd
- ▶ Power Supply: 5v, 5A $\Rightarrow$ 25W
 - ▶ Too much for some solutions :-(
- ▶ Reboots if power cycles - power surprisingly reliable on stage, but not at sound check
 - ▶ 90 seconds for reboot is an eternity if waiting for it to get sound check
 - ▶ Soundees... Unreliable drunks, except for rare and valuable exceptions.
 - ▶ Offer love, drugs, sex, money... In my experience nothing helps

Conclusion?

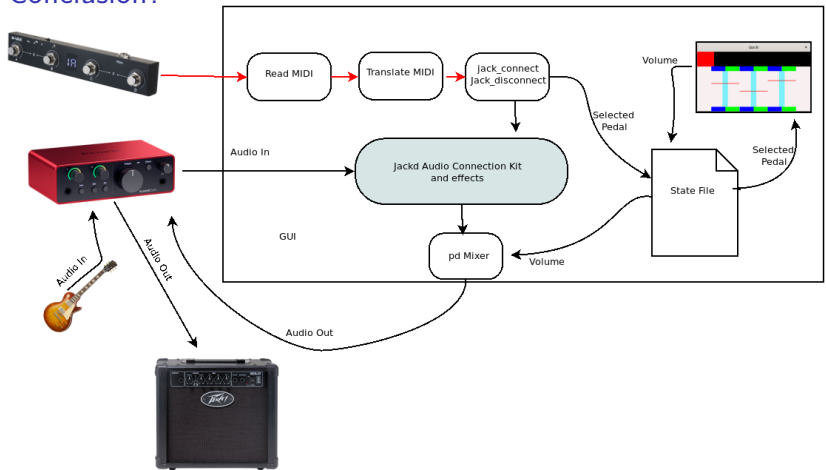


Figure 6: Block diagramme of complete system

The Future

An Ordered List

0. Health check
1. Watchdog/restart process
2. Banks
3. Temperature indicator
4. Tuner

`pandoc -t beamer -s Introduction.md -o Introduction.pdf`